Test Results

Freshness model on own photos

Test set: `data/own_test`, 27 photos (14 fresh, 13 rotten). Script: `evaluate_own.py`.

Check	Result
Overall accuracy	77.8% (21 of 27)
Fresh photos correct	11 of 14
Rotten photos correct	10 of 13

Confidence threshold sweep

Script: `scripts/threshold_sweep.py`. A photo is called fresh only if the fresh probability is at least t.

Threshold t	Fresh correct	Rotten correct	Overall
0.5	13/14	10/13	85.2%
0.6	12/14	10/13	81.5%
0.7	10/14	10/13	74.1%
0.8	10/14	10/13	74.1%
0.9	6/14	12/13	66.7%

Best threshold: 0.5 (85.2% overall).

"Not sure" band

With a middle band where the model answers "not sure":

Band	Answered	Correct
0.3 to 0.7	23 of 27	19 of 23
0.25 to 0.75	23 of 27	19 of 23
0.2 to 0.8	22 of 27	18 of 22

Errors seen at threshold 0.5

- Rotten photos called fresh: rotten_01 (0.86), rotten_03 (0.93), rotten_13 (0.83)
- Fresh photo called rotten: fresh_01 (fresh probability 0.07)

Bugs found and fixed

Bug	Fix
Backend crashed with `NameError: name 'app' is not defined` in `backend/main.py`	A router was registered before `app = FastAPI()`. Routers are now registered after the app is created
`.gitignore` was corrupted by NUL characters (written by a PowerShell command)	Rewritten as a clean UTF-8 file

Known issues

- Dishes missing from the training classes are mapped to the closest known dish (for example biryani shown as fried rice) until the model is fine-tuned on them.
- The freshness model sometimes calls rotten food fresh (3 of 13 rotten photos).
- Calorie values are estimates. Portion size and hidden ingredients affect real values.